Multisensor Fusion and Explicit Semantic Preserving-Based Deep Hashing for Cross-Modal Remote Sensing Image Retrieval

Yuxi Sun, Shanshan Feng, Yunming Ye, Xutao Li, Jian Kang, *Member, IEEE*, Zhichao Huang, and Chuyao Luo

***Abstract*— Cross-modal hashing is an important tool for retrieving useful information from very-high-resolution (VHR) optical images and synthetic aperture radar (SAR) images. Dealing with the intermodal discrepancies, including both spatial–spectral and visual semantic aspects, between VHR and SAR images is extremely vital to generate high-quality common hash codes in the Hamming space. However, existing cross-modal hashing methods ignore the spatial–spectral discrepancy when representing VHR and SAR images. Moreover, existing methods employ derived supervised signals, such as pairwise training images, to implicitly guide hashing learning, which fails to effectively deal with the visual semantic discrepancy, i.e., cannot adequately preserve the intraclass similarity and interclass discrimination between VHR and SAR images. To address these drawbacks, this article proposes a multisensor fusion and explicit semantic preserving-based deep Hashing method, termed as MsEspH, which can effectively deal with the discrepancies. Specifically, we design a novel cross-modal hashing network to eliminate the spatial–spectral discrepancies by fusing extra multispectral images (MSIs), which are generated in real time by a generative adversarial network. Then, we propose an explicit semantic preserving-based objective function by analyzing the connection between classification and hash learning. The objective function can preserve the intraclass similarity and interclass discrimination with class labels directly. Moreover, we theoretically verify that hash learning and classification can be unified into a learning framework under certain conditions. To evaluate our method, we construct and release a large-scale VHR-SAR image dataset. Extensive experiments on the dataset demonstrate that our method outperforms various state-of-the-art cross-modal hashing methods.**

***Index Terms*— Content-based remote sensing (RS) image retrieval, cross-modal dataset, cross-source image retrieval, deep cross-modal hashing, optical and synthetic aperture radar (SAR) data.**

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I. Introduction

With the increase in recent Earth observation (EO) satellite missions, big EO archives containing large-scale remote sensing (RS) images are produced [1]–[3]. As one of the fundamental techniques for mining crucial information from big EO archives, content-based RS image retrieval (CBRSIR) has attracted great attention in the RS community [4]–[9]. Learning to hash is a common solution for large-scale CBRSIR [10], which encodes RS images into compact hash codes by learning nonlinear hashing functions. Once an effective hashing function is learned, the original CBRSIR task can be transformed into a hash-code-based retrieval task.

Existing learning to hash algorithms can be categorized into unimodal hashing methods and cross-modal hashing methods. Unimodal methods [1], [10]–[18] only involve single modality satellites when learning hashing functions of RS images. In contrast, cross-modal methods [19], [20] use multiple modality satellites images to generate nonlinear hashing functions. Generally, cross-modal methods have more application scenarios than unimodal methods. This is because different modality satellites can provide complementary and correlated information, especially for optical and synthetic aperture radar (SAR) satellites [2]. For example, optical satellites can produce easy-to-understand images with rich texture information, but they only work well during daytime and fine weather. SAR satellites can achieve full-time and all-weather EOs, but their images are difficultly interpreted with fewer texture features, even for well-trained experts. Therefore, the cross-modal hashing between optical and SAR images can complement each other and operate well in many real-world situations.

Current optical and SAR cross-modal hashing methods mainly focus on medium-resolution optical images (MRIs) and rarely consider very-high-resolution (VHR) optical images. Usually, the cross-modal hashing between VHR and SAR images has a greater value than the cross-modal hashing between MRI and SAR images because, compared with MRIs, VHR images can accurately identify the spatial contours of small objects. For example, in maritime search and rescue surveillance [21], [22], target vessels are more likely to be found by the hashing methods between VHR and SAR images than the hashing methods between MRI and SAR images because VHR images can reveal the detailed structure and

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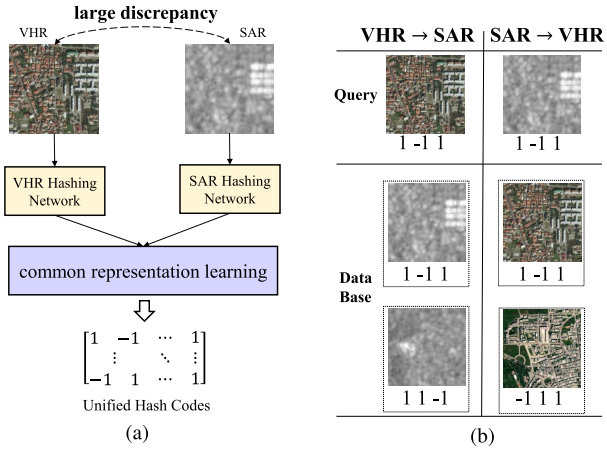


Fig. 1. Diagram of cross-modal hashing for VHR and SAR image retrievals. "VHR → SAR" denotes retrieving SAR images given a VHR image as a query. "SAR → VHR" denotes retrieving VHR images given an SAR image as a query. (a) Learning the hash codes of VHR and SAR images. (b) Cross-modal image retrieval-based learned hash codes.

diverse features of target vessels. In an emergency disaster assessment [23], VHR and SAR hashing retrieval may detect more damage regions than MRI and SAR hashing retrieval since VHR images can provide information at the scale of a single building. Overall, the cross-modal hashing methods between VHR and SAR images have a wide range of applications, and hence, we mainly study them in this article.

The problem of cross-modal hashing between VHR and SAR images is illustrated in Fig. 1. To solve this problem, we need to deal with intermodal discrepancies between them for learning the common hash codes in the Hamming space. Specifically, the intermodal discrepancies include two aspects: *spatial–spectral* and *visual semantic*. First, the spatial–spectral discrepancy refers to that VHR and SAR sensors detect different parts of the electromagnetic spectrum and capture different sizes of the smallest objects. Due to the large spatial–spectral discrepancy, the common hash codes between VHR and SAR images are not easily aligned in the Hamming space [20]. Second, the visual semantic discrepancy means that the visual characteristics of each class are vastly different between VHR and SAR images. For example, the VHR images of the residential class are more similar to the VHR images of the industrial class than the SAR images of the same class. Usually, the large visual semantic discrepancy makes it difficult to effectively learn the intraclass similarity and interclass discrimination between cross-modal images. Thus, the two discrepancies need to be handled during learning hash functions, especially for VHR and SAR images because their distinct imaging mechanisms lead to very large discrepancies.

However, existing cross-modal hashing methods [19], [20] cannot effectively deal with the discrepancies when applied directly to VHR and SAR image retrievals for two main reasons. First, existing methods only consider the spatial-spectrum information of VHR and SAR themselves when designing hashing networks to represent them. This makes the spatial–spectral discrepancy cannot be effectively eliminated. Second, existing methods usually employ derived supervision signals, such as pairwise training images, to train hashing networks. These supervision signals cannot explicitly preserve the intraclass similarity and the interclass discrimination between VHR and SAR images. Thus, the visual semantic discrepancy is also unable to be effectively dealt with.

To address these limitations, we propose a multisensor fusion and explicit semantic preserving-based deep Hashing method, termed **MsEspH**, for VHR and SAR image retrievals. MsEspH contains a cross-modal hashing network and an objective function. The cross-modal network consists of a VHR hashing network and an SAR hashing network that are utilized to compact VHR and SAR images into hash codes, respectively. The objective function is used to train the two hashing networks and constrain them into a common semantic space.

To eliminate the spatial–spectral discrepancy, we propose a novel VHR hashing network, which can fuse extra multispectral image (MSI) information. The MSIs cover the visible and infrared portions of the electromagnetic spectrum and can provide supplementary data for VHR images [24], [25]. Thus, when learning common hash codes, fusing MSIs is able to bridge the spatial–spectral gap between VHR and SAR images. Specifically, we adopt a Pix2Pix model [26] to generate MSIs and use a multisensor fusion module to combine the MSIs, where the Pix2Pix is an image-to-image translation network. By fusing real-time MSIs in this way, the spatial–spectral discrepancy can be effectively reduced.

To deal with the visual semantic discrepancy, we design an explicit semantic preserving objective function by analyzing the connection between classification and hash learning. The objective function is derived from a softmax loss and can directly use class labels to guide hashing networks. This allows the generated hash codes to preserve the intraclass similarity and the interclass discrimination as much as possible. Moreover, we prove that hash learning and classification can be unified into a learning framework.

To sum up, our contributions can be listed as follows.

1) We study a novel research problem, which learns cross-modal deep hashing methods for the retrieval task of VHR and SAR images. To the best of our knowledge, this is the first work to investigate generating high-quality hash codes for VHR and SAR images.
2) We develop a new deep hashing method, namely, MsEspH, to jointly produce hashing representations for VHR and SAR images. We design a novel cross-modal hashing network to reduce the spatial–spectral discrepancy by a multisensor fusion module combining extra MSIs. Moreover, we propose a novel objective function to explicitly preserve the intraclass similarity and the interclass discrimination between VHR and SAR images. Meanwhile, we also prove that classification and hash learning can be integrated into a unified framework.
3) We conduct extensive experiments on a VHR-SAR image dataset, and the results show that our method consistently outperforms state-of-the-art cross-modal hashing methods.
4) To the best of our knowledge, there are no existing cross-modal datasets containing VHR and SAR images. Hence, we collect and build a large-scale benchmark

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dataset, named CBRSIR_VS. We make it publicly available to facilitate the study of VHR and SAR images.

The rest of this article is organized as follows. Section II briefly reviews related work. Section III presents our method in detail. Section IV introduces our released dataset and the experimental results in detail. We make a summary of this article in Section V.

II. Related Work

In this section, we mainly conduct a literature review for unimodal hashing methods and cross-modal hashing methods.

A. Unimodal Hashing Methods

Unimodal hashing methods [1], [10]–[18] learn hashing functions to compact RS images into binary code by preserving the similarity between intramodality RS images produced by the same satellite sensor. Early studies, such as [1] and [11], known as shallow hashing methods, focus on taking handcrafted features [11], [27], [28] as input and learning some kernel functions or decision trees to generate hash codes. Generally, shallow methods cannot generate highly discriminative hash codes due to the limited representation ability of handcrafted features.

To capture the rich semantic information of RS images, some deep learning methods [10], [12]–[18] take convolutional neural networks to jointly learn representation and hash coding. For example, several works [10], [29] construct pairwise or triplewise training images based on class labels and adopt convolutional neural networks to learn hashing functions. The construction method of training images is similar to deep metric learning [30], which can learn the fine-grained semantic relationship between RS images. Song *et al.* [31] fuse the similarity information of image pairs and semantic information of class labels to generate more discriminative hash codes. Besides, to reduce storage space and computing resources, Li *et al.* [18] propose a lightweight deep neural network for effective RS image hashing. Shan *et al.* [32] present a proxy-based hash retrieval method that combines hash code learning with proxy-based metric learning in a convolutional neural network. However, these methods are unimodal approaches and fail to handle the real-life multimodal scenarios, such as cross-modal retrieval [33] and multimodal named entity recognition [34]. Specifically, in many practical applications, such as all-weather rescue [35], cross-modal images are required and the unimodal methods cannot be directly applied since they cannot learn cross-modal hashing functions. Hence, cross-modal methods [19], [20] get more and more attention.

B. Cross-Modal Hashing Methods

Cross-modal hashing methods [19], [20], [36]–[43] learn hashing functions to compact cross-modal data into hash codes in a common Hamming space, where query data and database data are in different modalities. The core of cross-modal retrieval is how to represent data and measure the content similarity between different types of data. In particular, feature representation is the basis of many deep learning-based methods, such as RS image interpretation [3] and deep learning-based cloud detection [44]. In the section, we review related methods in the general nature image domain and the RS image domain. In the section, we review related methods in the general nature image domain and RS image domain.

1) Nature Image Domain: Cross-modal hashing methods [36]–[43], including traditional methods [36]–[39] and deep learning methods [40]–[43], mainly focus on text and images in the nature image domain. Cross-modal related tasks are usually very challenging because distinct modal information needs to be considered at the same time [34], [43]. For example, Song *et al.* [36] design a novel intermedia hashing model to explore the correlations among multiple media types from different data sources. Ding *et al.* [38] employ a collective matrix factorization technology to learn cross-modal hash functions. Zhang and Li [39] seamlessly integrate semantic labels into a cross-modal hashing learning procedure and can use supervised information with linear-time complexity. Jiang and Li [40] integrate feature learning and hash-code learning into an end-to-end deep learning framework for text and image retrieval. Jin *et al.* [43] design a cross-modal hashing model by exploring the correlation structure and semantic information between different modalities.

2) Remote Sensing Image Domain: Cross-modal retrieval in the RS image domain mainly contains image–image retrieval [45]–[48], audio–image retrieval [49]–[51], and text–image retrieval [33]. Cross-modal hashing is a popular solution for cross-modal retrieval. For example, Li *et al.* [19] propose a new source-invariant deep hashing network for panchromatic and MSI retrieval. Xiong *et al.* [20] design a cross-modal hashing network with a new training strategy to eliminate the large cross-modality variations for optical and SAR image retrieval.

Existing cross-modal hashing methods [19], [20] usually fail when applied directly to VHR and SAR images for the following reasons. First, when representing RS images, the spatial-spectrum discrepancy cannot be effectively eliminated because current methods only consider the spatial-spectrum information of these images themselves. Second, existing methods fail to directly deal with the visual semantic discrepancy by class labels. This is because they generally utilize derived supervised signals, such as pairwise training images, to train hashing networks, and these supervision signals cannot explicitly preserve the intraclass and interclass similarities. To address these limitations, we propose a multisensor fusion and explicit semantic preserving-based deep cross-modal hashing method.

III. Method

A. Notation and Task Definition

In the task of cross-modal RS image retrieval, we are given the n pairs of the VHR-SAR image dataset $\mathcal{O} = \{o_i\}_{i=1}^{n}$, and $o_i = (x_i, y_i, c_i)$, where x_i and y_i represent a VHR image and an SAR image, respectively. $c_i \in \{1, 2, \ldots, l\}$ denotes the class label of o_i, where l is the total number of class labels. The goal of cross-modal hashing for VHR and SAR image retrieval is to compact x and y into K-bit hash codes by learning two nonlinear hashing functions: $f_x : x \mapsto h^x \in \{-1, 1\}^K$ and

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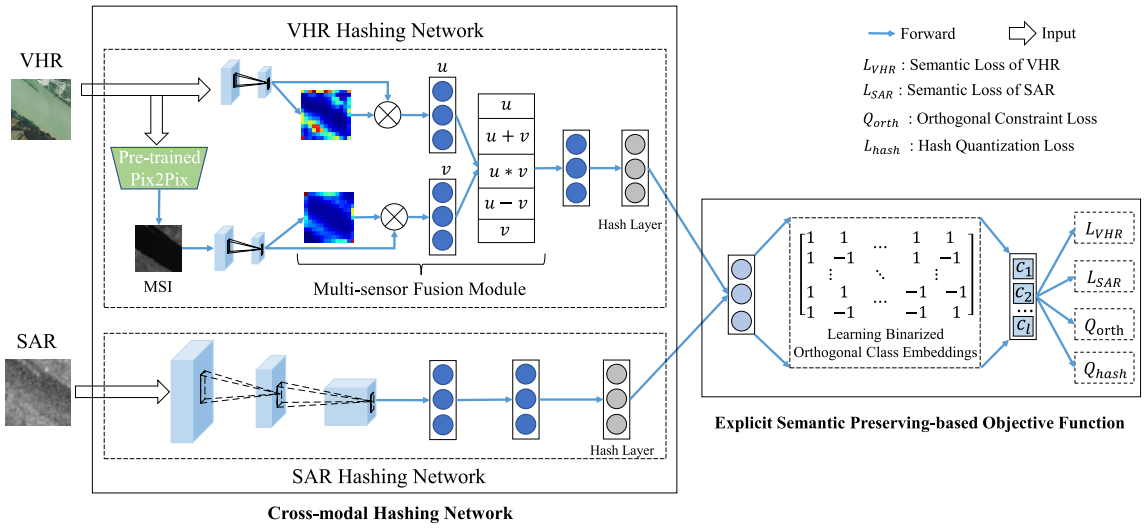


Fig. 2. Overview of MsEspH, which contains a cross-modal hashing network and an explicit semantic preserving-based objective function. The cross-modal hashing network consists of a VHR optical image hashing network and an SAR image hashing network. In the VHR hashing network, the pretrained Pix2Pix is an adversarial network, which translates a VHR image into a corresponding MSI.

$f_y : y \mapsto h^y \in \{-1, 1\}^K$. The hashing functions need to enforce h^x and h^y to have a small (large) Hamming distance when x and y belong to the same class (different classes).

B. Overview

As shown in Fig. 2, we propose a multisensor fusion and explicit semantic preserving-based deep hashing method, called **MsEspH**, which mainly contains two components: a new cross-modal hashing network and an explicit semantic preserving-based objective function.

The cross-modal hashing network consists of a VHR hashing network and an SAR hashing network, which are used to generate hash codes of VHR and SAR images, respectively. Specifically, the VHR hashing network contains a pretrained Pix2Pix model [26], a convolutional layer, a multisensor fusion module, and a hash layer. To generate MSIs in real time, the Pix2Pix is applied. The convolutional layer is similar to an AlexNet [52] and is employed to extract the features of MSI and VHR images. The multisensor fusion module is designed to combine these RS images to generate the final representation of VHR images. The hash layer adopts tanh functions to generate binary hash codes. The SAR hashing network is comprised of a convolution network adopted AlexNet architecture, a full connection layer, and a hash layer, which also uses the tanh activation function.

The explicit semantic preserving-based objective function is used to constrain the two hashing networks into a common semantic space. The novel objective function is derived from a softmax function. In addition, we analyze the relationship between hashing learning and classification. Then, the objective function is designed by reconstructing the softmax function with certain conditions, such as the binarization constraint of class embeddings. The objective function unifies hashing learning and classification into an end-to-end framework, which can explicitly preserve the intraclass similarity and the interclass discrimination with class labels.

C. Cross-Modal Hashing Network

1) VHR Hashing Network: The VHR images have a large spatial–spectral discrepancy with SAR images due to the difference in imaging techniques. For example, a VHR sensor detects very high-frequency radiation within the visible section of the electromagnetic spectrum. In contrast, an SAR sensor operates at much lower frequencies within the microwave section of the spectrum [53]. The spectral difference obstructs the alignment of common hash codes in the Hamming space. Besides, different spatial information between VHR and SAR images also aggravates the problem. Thus, when representing VHR images, we consider utilizing additional sensor data, i.e., multispectral sensor data, to reduce the spatial–spectral discrepancy. MSIs cover the visible and infrared portions of the electromagnetic spectrum, such as near-infrared and shortwave infrared, and have various spatial resolutions, which can provide supplementary spectral and spatial information for VHR images. Fusing MSIs, the visual image interpretation of VHR images can be improved effectively [24], [25], [54], which is extremely useful to reduce the spatial–spectral discrepancy. Moreover, compared with SAR images, VHR images are easier to fuse with MSI images [55].

To enable the VHR hashing network to obtain MSIs in real time, we train an image translation model of VHR-to-MSI using a Pix2Pix network. The pretrained Pix2Pix model can be applied to generate MSIs, which provides supplementary information when learning the representation of VHR images. The Pix2Pix network is a conditional adversarial network that processes image-to-image translation problems.

a) VHR-to-MSI: The Pix2Pix learns a mapping from a VHR image x to an MSI x^M, $G: x \mapsto x^M$, where G is called a generator, x^M represents the MSI that has the same geolocation with x, and the image-pair (x, x^M) is constructed and aligned based on geolocation before training. The generator G is trained to generate "fake" MSI that cannot be distinguished from "real" MSI by a discriminator D, which is adversarially

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trained to detect the generator's "fakes" and possible. The objective function can be expressed as

$$G^* = \arg\min_G \max_D \mathcal{L}_{\text{pix2pix}}(G, D) + \lambda \mathcal{L}_{L1}(G) \tag{1}$$

where $\mathcal{L}_{\text{pix2pix}}(G, D)$ is an adversarial loss and $\mathcal{L}_{L1}(G)$ is an L1 loss.

The adversarial loss is employed to train generator G and discriminator D, which is defined as follows:

$$\mathcal{L}_{\text{pix2pix}}(G, D) = \mathbb{E}_{x,x^M}\left[\log D\left(x, x^M\right)\right] + \mathbb{E}_x\left[\log(1 - D(x, G(x)))\right] \tag{2}$$

where G is trained to minimize the loss against an adversarial D that tries to maximize it. In addition to the adversarial loss, the Pix2Pix also uses the L1 loss to reduce image blurring. The L1 loss indicates a penalized loss, which is defined as follows:

$$\mathcal{L}_{L1} = \mathbb{E}_{x,x^M}\left[\|x^M - G(x)\|_1\right] \tag{3}$$

where $\|\cdot\|_1$ denotes an L1 norm. Based on (1)–(3), the Pix2Pix can be trained with aligned VHR-MSI image pairs. When learning the hashing function of VHR, the generator G of a pretrained Pix2Pix is used to generate an MSI based on a VHR, as shown in Fig. 2.

b) Multisensor fusion: To integrate the information of MSI and VHR images, we design a multisensor fusion module. Prior to it, we adopt an Alexnet network to extract the convolution features of these images before being processed by an average pooling. The convolution features of VHR images are denoted by $g_i \in \mathbb{R}^d, i = 1, \ldots, m$, where d indicates a feature dimension and m is the number of convolution kernels. Similar to VHR images, the convolution features of MSI are represented by $g_i^M \in \mathbb{R}^d, i = 1, \ldots, m$. RS images contain some useless background, which may have negative effects on the fusion performance. Hence, an attention module is designed to focus on key regions when generating an embedding by the multisensor fusion. It is defined as follows:

$$u = \sum_{i=1}^{m} \text{score}(g_i; \theta) \cdot g_i \tag{4}$$

$$v = \sum_{i=1}^{m} \text{score}\left(g_i^M; \theta_M\right) \cdot g_i^M \tag{5}$$

where u and v are the embedding of MSI and VHR images, $\text{score}(\cdot\,;\,\cdot)$ represents an attention score function. The core of the attention module is to learn a score function for each feature. The score function is designed by using a two-layer CNN with a softplus activation function at the top. For simplicity, we employ the convolutional filters of size 1×1, which works well in practice. The score function can output the relevance of image features.

Once the image embedding is generated, inspired by Conneau *et al.* [56], four fusion methods are applied to combine u and v into a single vector: 1) concatenation of the two representations (u, v); 2) elementwise add $u + v$; 3) elementwise product $u * v$; and 4) elementwise difference $u - v$. The fusion function is defined as follows:

$$\text{fusion}(u, v) = \text{concat}(u, v, u + v, u * v, u - v). \tag{6}$$

Based on the fusion function, we obtain a resulting vector, which captures information from both VHR and MSI. Then, the resulting vector is fed into a hash layer that outputs the hash codes of VHR, i.e., h^x.

2) SAR Hashing Network: Usually, a hashing network consists of a feature extractor, some fully connected layers, and a hash layer [57], [58]. Similar to [57] and [58], the SAR hashing network utilizes a classic hashing network architecture, which is composed of five convolutional layers, four pooling layers, two fully connected layers, and a hash layer, as shown in Fig. 2. The convolutional layers and pooling layers form the feature extractor. The hash layer is implemented by a tanh function. In addition, when time consumption is allowed to be increased, the feature extractor can be replaced by more sophisticated network architectures. In other words, our method is compatible with any CNN-based feature extractor.

To generate a K-dimensional feature vector of an SAR image, the SAR hashing network first extracts image features of y and applies a full connection layer to change the feature dimension. The K-dimensional vector is subsequently fed into a hash layer implemented with a tanh activation function to produce the hash vector of the SAR image, i.e., h^y.

To map h^x and h^y into a common semantic space, we design a novel objective function to enforce them to have a similar distribution when they share at least one common label. Moreover, the objective function can directly employ class labels to preserve the intraclass and interclass similarity between VHR and SAR images.

D. Explicit Semantic Preserving-Based Objective Function

The cross-modal hashing requires to effectively deal with the visual semantic discrepancy, i.e., preserve the intraclass and interclass similarities, between VHR and SAR images when mapping these cross-modal images into hash codes in common Hamming space [19], [20]. Due to the distinct imaging techniques and acquisition geometries between VHR and SAR images, the visual semantic discrepancy is rather large, and the intraclass and interclass similarities are not easy to be learned. Therefore, we directly use class labels as supervision signals to fully capture the complex semantic of cross-modal RS images. To achieve the goal, we comprehensively analyze the connection between classification and hashing learning. Then, an explicit semantic preserving-based objective function is designed, which derives from the classification task. It can guide hashing learning using class labels directly. In this section, we describe the process of the objective function in detail.

1) Connection of Classification and Hash Learning: The classification tasks usually adopt a softmax function to train a model. The softmax function outputs a vector that represents the probability distributions of multiclass labels, which is defined as follows:

$$L = \frac{1}{n}\sum_{i=1}^{n} -\log p_i = \frac{1}{n}\sum_{i=1}^{n} -\log \frac{\exp\left(f_j\right)}{\sum_k \exp(f_k)} \tag{7}$$

where p_i is the normalized probability of the ith sample, and f_j denotes the jth element ($j \in \{1, 2, \ldots, l\}$ indicates the

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jth class) of the vector of class scores $\boldsymbol{f}$. In the softmax loss, $\boldsymbol{f}$ usually is the product between a class embedding matrix $\mathbf{E}$ and a feature vector $\mathbf{h}$, i.e., $\boldsymbol{f} = \mathbf{E}^T\mathbf{h}$, where $\mathbf{E} \in \mathbb{R}^{K\times l}$. In our method, $\mathbf{h}$ is the output of the VHR hashing network or the SAR hashing network. $\sum_k \exp(f_k)$ is the sum score of all classes and can be divided into two parts: the score of a target class and the sum score of the other classes. Thus, $\log p_i$ can be denoted as follows:

$$\begin{aligned}\log p_i &= \log \frac{\exp\left(E_j^T h_i\right)}{\sum_k \exp\left(E_k{}^T h_i\right)} \\ &= \log \frac{\exp\left(E_j^T h_i\right)}{\exp\left(E_j{}^T h_i\right) + \sum_{k!=j} \exp\left(E_k{}^T h_i\right)}\end{aligned} \tag{8}$$

where E_j is the jth column of $\mathbf{E}$ and h_i represents h_i^x or h_i^y. Based on (8), if $\sum_{k!=j} \exp(E_k{}^T h_i)$ is approximately a constant, we can obtain the positive correlation between $\log p_i$ and $E_j^T h_i$, which is described as follows:

$$\begin{aligned}\max \log p_i &= \max \log \frac{\exp\left(E_j^T h_i\right)}{\exp\left(E_j{}^T h_i\right) + \sum_{k!=j} \exp\left(E_k{}^T h_i\right)} \\ &\approx \max \log \frac{\exp\left(E_j^T h_i\right)}{\exp\left(E_j{}^T h_i\right) + \text{const}} \\ &\propto \max E_j^T h_i.\end{aligned} \tag{9}$$

From (9), we can learn that maximizing $\log p_i$ is approximately to maximize $E_j^T h_i$. Besides, according to (7), optimizing the softmax loss is equivalent to maximizing $\log p_i$. Therefore, the optimization of classification can be approximately transformed to maximize $E_j^T h_i$. Furthermore, according to Zhu *et al.* [57], we find that the Hamming distance $\text{dist}_H(\cdot,\cdot)$ has a linear relationship with inner product $\langle\cdot,\cdot\rangle$, which is defined as follows:

$$\text{dist}_H\left(E_j, h_i\right) = \frac{1}{2}\left(K - \langle E_j, h_i\rangle\right) \tag{10}$$

where K is the hash code length, and $E_j \in \{-1, 1\}^K$. Based on (7), (9), and (10), the softmax loss can be represented as follows:

$$\begin{aligned}\max \log p_i &\propto \max E_j^T h_i \\ &= \min \frac{1}{2}\left(K - E_j^T h_i\right) \\ &= \min \text{dist}_H\left(E_j, h_i\right).\end{aligned} \tag{11}$$

According to (11), maximizing $\log p_i$ is equivalent to optimizing the Hamming distance between hash codes and class labels. This means that hash codes can preserve the class similarities by (11), which is the target of hashing learning. Hence, classification and hash learning can be optimized simultaneously, and there is a strong connection between these two tasks. In other words, the softmax loss can optimize classification; meanwhile, it can also encourage the hash codes of similar VHR-SAR image pairs to be close to a common class embedding.

As shown in Fig. 3, the VHR image h_1^x and the SAR image h_1^y belong to class one. To preserve the intraclass similarity between VHR and SAR images, our proposed objective function enforces h_1^x and h_1^y to be distributed around class vector E_1. Meanwhile, E_1 is constrained to be a binarized vector. In this way, h_1^x and h_1^y can be compacted into the same hash codes, which effectively eliminates the intermodal discrepancy. Besides, the VHR image h_3^x and the SAR image h_3^y do not belong to class one. The objective function would increase the Hamming distance between E_1 and $h_3^*, * \in \{x, y\}$. This also means that interclass discrimination is maintained. In a word, the objective function can unify classification and hash learning into a learning framework to explicitly preserve intraclass similarity and interclass discrimination between VHR and SAR images.

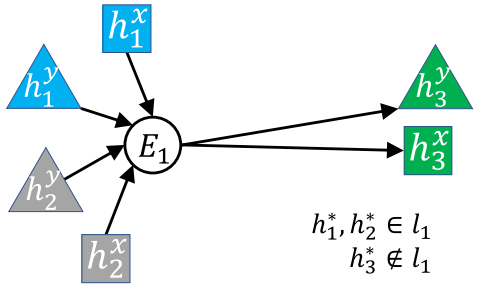


Fig. 3. Example of the connection relationship between classification and hash learning. Squares represent VHR images, and triangles indicate SAR images. h_i^x and h_i^y, $i \in [1, 3]$, denote the hash codes of a VHR image and an SAR image, respectively. h_i^* means h_i^x or h_i^y. h_1^* and h_2^* belong to class one, and h_3^* does not belong to class one. E_1 is the embedding vector of class one. As shown in Fig. 3, our proposed objective function constrains E_1 as a binarized vector, meanwhile, enforces h_1^* and h_2^* to be close to E_1, and keeps h_3^* away from E_1.

2) Explicit Semantic Preserving Loss: As mentioned above, when $\sum_{k!=j} \exp(E_k{}^T h_i) \propto$ const and $E_* \in \{-1, 1\}^K$, hash learning can be jointly optimized with classification. Based on the conclusion, we design a new objective function.

To make $\sum_{k!=j} \exp(E_k{}^T h_i)$ approximate to a constant, we let h_i be orthogonal to $E_k (k \neq j)$, which results

$$\sum_{k!=j} \exp\left(E_k{}^T h_i\right) \propto 0. \tag{12}$$

Afterward, according to (9), the optimal solution of classification can be approximately obtained by maximizing $E_j^T h_i$. The optimal solution indicates that E_j and h_i are parallel and in the same direction. Based on this, we can get the following results:

$$E_j \perp E_k, k \in \{1, 2, \ldots, l\} \wedge k \neq j \tag{13}$$

where j indicates a certain class. The classification task needs to optimize all class labels, which means that the columns of $\mathbf{E}$ are orthogonal to each other. Besides, E_* is a binarized vector according to (10), and h_i represents h_i^x or h_i^y.

Thus, based on (7), (9), (10), and (13), we propose an explicit semantic preserving-based loss to train `MsEspH` with class labels directly, which is defined as follows:

$$\min_{\Theta} \mathcal{L}_{\text{MsEspH}} = L_{\text{VHR}} + L_{\text{SAR}} + \alpha Q_{\text{orth}} + \beta Q_{\text{hash}} \tag{14}$$

where L_{VHR} is the semantic loss of VHR, L_{SAR} is the semantic loss of SAR, Q_{orth} is the orthogonal constraint loss of E_*, and Q_{hash} is the hash quantization loss of E_*. Besides, Θ represents the whole parameter space of `MsEspH`, and α and β are two

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hyperparameters. Based on $\mathcal{L}_{\text{MsEspH}}$, our proposed network can be optimized to learn hashing functions. The impact of the two hyperparameters will be discussed in the experiments. The detailed configuration of the proposed network and experimental results will be discussed in Section IV. Next, we present the four parts of loss in (14).

L_{VHR} and L_{SAR} learn the semantic structure between VHR and SAR images, and these images can be compacted into hash codes with shared semantics in the Hamming space. Moreover, these two loss functions can explicitly preserve the intraclass similarity and the interclass discrimination between VHR and SAR mages. L_{VHR} and L_{SAR} are defined as follows:

$$L_{\text{VHR}} = \frac{1}{n}\sum_{i=1}^{n} -\log \frac{\exp\left(E_j^T h_i^x\right)}{\sum_k \exp\left(E_k{}^T h_i^x\right)} \tag{15}$$

$$L_{\text{SAR}} = \frac{1}{n}\sum_{i=1}^{n} -\log \frac{\exp\left(E_j^T h_i^y\right)}{\sum_k \exp\left(E_k{}^T h_i^y\right)}. \tag{16}$$

Q_{orth} needs to enforce the class embeddings to be orthogonal, and two vectors can be constrained to be orthogonal by making their inner product close to zero. Thus, Q_{orth} can be defined as follows:

$$Q_{\text{orth}} = \sum_{j=1}^{m}\sum_{k!=j} \| E_j^T E_k - \mathbf{I} \|_2 \tag{17}$$

where $\mathbf{I}$ represents an identity matrix. As described in [58], when the binarized vectors are orthogonal, the Hamming distances between these vectors are quite large. Thus, with (17), interclass discrimination is maintained, and the generated hash codes of different classes become easily distinguishable.

Q_{hash} requires enforcing the class embeddings to binarized vectors, which can be defined as follows:

$$Q_{\text{hash}} = \sum_{i=1}^{l} \| \, |\mathbf{E}_i| - \mathbf{1} \, \|_1 \tag{18}$$

where $|\cdot|$ calculates the elementwise absolute value and $\|\cdot\|_1$ means L1 normalization. Equation (18) can be used to control the quantization error between continuous features and binarized features. Since Q_{hash} is nonsmooth and its derivative cannot be computed, we adopt a smoothing surrogate of the absolute function [59]: $|x| \approx \log\cosh x$, which reduces (18) as follows:

$$Q_{\text{hash}} = \sum_{i=1}^{l}\sum_{t=1}^{K} \log\cosh(|\mathbf{E}_{it}| - \mathbf{1}). \tag{19}$$

IV. Experiments

A. Experimental Setup

1) Dataset: In our experiments, we need a dataset of VHR and SAR images to evaluate the performance of MsEspH. To the best of our knowledge, there is no open cross-modality dataset for VHR and SAR image retrieval. Thus, we construct a VHR-SAR dual-modality RS image dataset by using the class labels and geolocation of EuroSAT [60], which is a unimodal multispectral RS image dataset. The VHR-SAR image dataset is named **CBRSIR_VS**, which contains ten class labels and 26 901 pairs of VHR and SAR images. Different from EuroSAT, CBRSIR_VS is a cross-modal dataset and consists of VHR and SAR images.

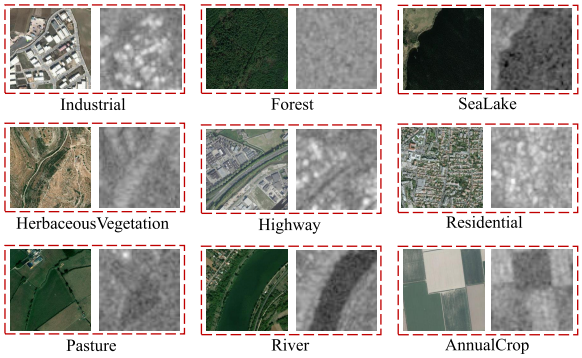


Fig. 4. Some pairs of VHR and SAR images in CBRSIR_VS. In each pair, the left one is the VHR image, and the right one is the SAR image.

In CBRSIR_VS, VHR images are collected from Google Earth Imagery and ArcGIS map [61], and SAR images are collected from Sentinel-1 imagery via Google Earth Engine [62], a cloud computing-based RS platform. When constructing the dataset, we need to keep the imaging data consistency across different data modalities. As we know, Google Earth can display the imaging date of a satellite image when the corresponding latitude and longitude are provided. ArcGIS also has a similar tool. Thus, we first download VHR images and save their imaging dates. Then, based on these imaging dates, for each VHR image, an SAR image with a close timestamp is extracted via Google Earth Engine. Finally, in this way, the imaging dates of VHR and SAR images are kept consistent. Besides, when collecting VHR and SAR images, we choose the same geographic locations with the EuroSAT [60] dataset, in which most of the regions are extracted from nonurban areas, and the change rate of land use is relatively small. In these regions, even though the imaging dates of VHR and SAR images are slightly different, their class labels are regarded as the same. Thus, the cross-modal RS images can be considered consistent, and we are able to conduct the cross-modal retrieval experiments on the dataset.

In particular, each VHR image is of size 256*256 pixels and has three spectral bands, i.e., red, blue, and green bands, with a spatial resolution of 1 m. Same with [63], SAR images are restricted to vertically polarized (VV) data with the interferometric wide (IW) swath mode, and we ignore other polarizations. The size of each SAR image is 64*64 pixels with a spatial resolution of 10 m. Some examples of VHR and SAR images are shown in Fig. 4. For this dataset, we randomly select 100 images per class as the test set, and the remaining images are used as the retrieval set. We further randomly select 100 images per class from the retrieval set as the validation set and treat the others as the training set. The CBRSIR_VS dataset is publicly available.[1]

2) Baselines: We compare MsEspH with nine supervised cross-modal hashing methods: four traditional shallow cross-modal hashing methods (IMH [36], LSSH [37],

[1]https://github.com/sunyuxi/CBRSIRVS

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TABLE I
COMPARISON IN MAP OF THE HAMMING RANKING FOR DIFFERENT BITS ON IMAGE RETRIEVAL

Architecture	Method	VHR → SAR				SAR → VHR			
		16 bits	24 bits	32 bits	48 bits	16 bits	24 bits	32 bits	48 bits
Shallow	IMH	0.1193	0.1478	0.1588	0.1600	0.1324	0.1728	0.1851	0.1929
	LSSH	0.2416	0.2563	0.2648	0.2660	0.2224	0.2466	0.2532	0.2732
	CMFH	0.2657	0.2461	0.2944	0.3096	0.2571	0.2665	0.3016	0.3226
	SCM	0.4735	0.4958	0.5163	0.5107	0.4765	0.4813	0.4923	0.5042
Deep	DCMHN	0.6189	0.6257	0.6312	0.6368	0.6537	0.6850	0.6924	0.6996
	CMHH	0.7432	0.7486	0.7532	0.7557	0.7734	0.7732	0.7847	0.7918
	PRDH	0.7259	0.7522	0.7637	0.7830	0.7900	0.7943	0.8030	0.8079
	DCMH	0.7631	0.7917	0.7947	0.7920	0.7362	0.7434	0.7495	0.7563
	SIDHCN	0.7963	0.8119	0.8221	0.8396	0.7973	0.8035	0.8044	0.8051
	DSMHN	0.8857	0.8872	0.8906	0.8966	0.7974	0.8037	0.8042	0.8058
	MsEspH	**0.9145**	**0.9314**	**0.9324**	**0.9398**	**0.8333**	**0.8344**	**0.8442**	**0.8458**

CMFH [38], and SCM [39]) and five recently proposed deep cross-modal hashing methods (DCMHN [20], CMHH [41], PRDH [42], DCMH [40], SIDHCNN [19], and DSMHN [43]). For the shallow learning-based methods, we adopt Alexnet [52], which has been trained on the ImageNet dataset [64], to extract deep features as the inputs. For the deep learning-based methods, we directly use raw image pixels as input. To make a fair comparison, the Alexnet is utilized as the same backbone network by all deep baseline methods. In particular, the Alexnet needs a three-channel input image, and the number of spectrum channels of SAR is one, which is not consistent with the number of the channels of the input image. Thus, SAR images are triplicated to generate a three-channel image. Consequently, all deep learning-based methods can directly deal with VHR and SAR images.

3) Implementation Details: MsEspH is implemented with Pytorch framework. We adopt stochastic gradient descent (SGD) with a minibatch size of 256 as our optimization algorithm. The learning rate of SGD optimizer is initialized to $10^{-1.5}$ and is linearly decreased to 10^{-6} with the grow of epoch number. MsEspH only involves two hyperparameters, i.e., α and β. We empirically set $\alpha = 1$ and $\beta = 1$. The hyperparameters' sensitivity is discussed in Section IV-G.

4) Evaluation Protocol: For a cross-modal hashing retrieval between VHR and SAR images, we generally conduct studies for two typical tasks: "VHR → SAR" (given a VHR image, retrieve the corresponding image in SAR database) and "SAR → VHR" (given an SAR image, retrieve the corresponding image in VHR database). In our experiments, the ground-truth RS images of a query are defined as the images that share at least one common label with the query. The cross-modal images are first mapped into hash codes, and then, the similarity is measured based on the Hamming distance. Finally, some common evaluation metrics are applied based on the similarity.

To evaluate the performance of the proposed MsEspH method, we adopt three widely used evaluation metrics: mean of average precision (mAP), TopN-precision (P@N), and precision–recall curves (PR curve). mAP is one of the most widely used criteria for evaluating retrieval accuracy [65], [66]. Given a query set, mAP is defined as an average of APs for all queries, where AP is the average precision of each query. AP is defined as follows:

$$\mathrm{AP} = \frac{1}{N}\sum_{r=1}^{R} P(r)\theta(r) \tag{20}$$

where N is the number of ground-truth instances in the database for each query and $P(r)$ is the precision of top r return results. $\theta(r) = 1$ if the rth returned result shares at least one common label with the query and $\theta(r) = 0$ otherwise. In our experiments, R is set to 1900. P@N denotes the precision with different numbers of retrieval instances. PR curves show the precision with different recall values. The larger area under the PR curve, the better retrieval performance.

B. Experimental Results

1) Comparison With Baseline Methods: Table I represents the mAP results for all methods on CBRSIR_VS with hash code lengths varying from 16 to 48. From the results, we can observe that shallow hashing methods obtain the worst performance even though they take the deep features as input. This is because shallow methods separately learn feature representations and hashing functions of cross-modal RS images, which limits the discrimination capability of generated hash codes. In contrast, deep hashing methods take convolutional neural networks as a feature representation layer and design a hashing layer to jointly perform representation learning and hash coding. As a result, in Table I, all deep methods are superior to shallow methods. Especially, MsEspH yields better performance than all baseline methods. For example, compared with DSMHN for "VHR → SAR," MsEspH achieves 3.25%, 4.98%, 4.69%, and 4.82% increments on 16-, 24-, 32-, and 48-bit hash codes, respectively. Moreover, compared with DSMHN for "SAR → VHR," MsEspH also achieves

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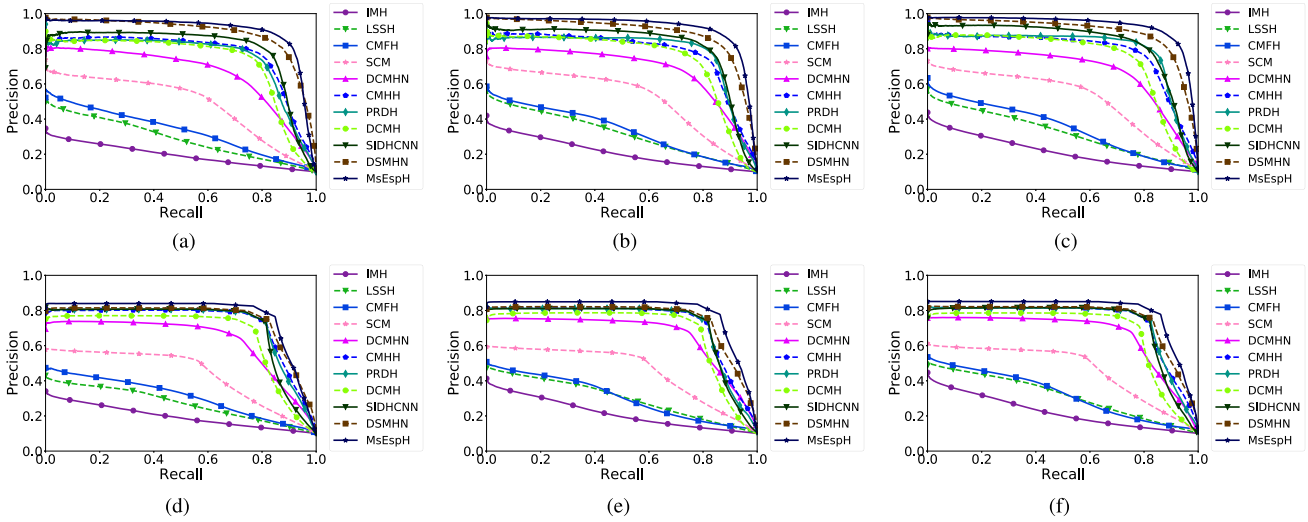


Fig. 5. PR curve on the CBRSIR_VS dataset. (a) VHR → SAR@16 bits. (b) VHR → SAR@32 bits. (c) VHR → SAR@48 bits. (d) SAR → VHR@16 bits. (e) SAR → VHR@32 bits. (f) SAR → VHR@48 bits.

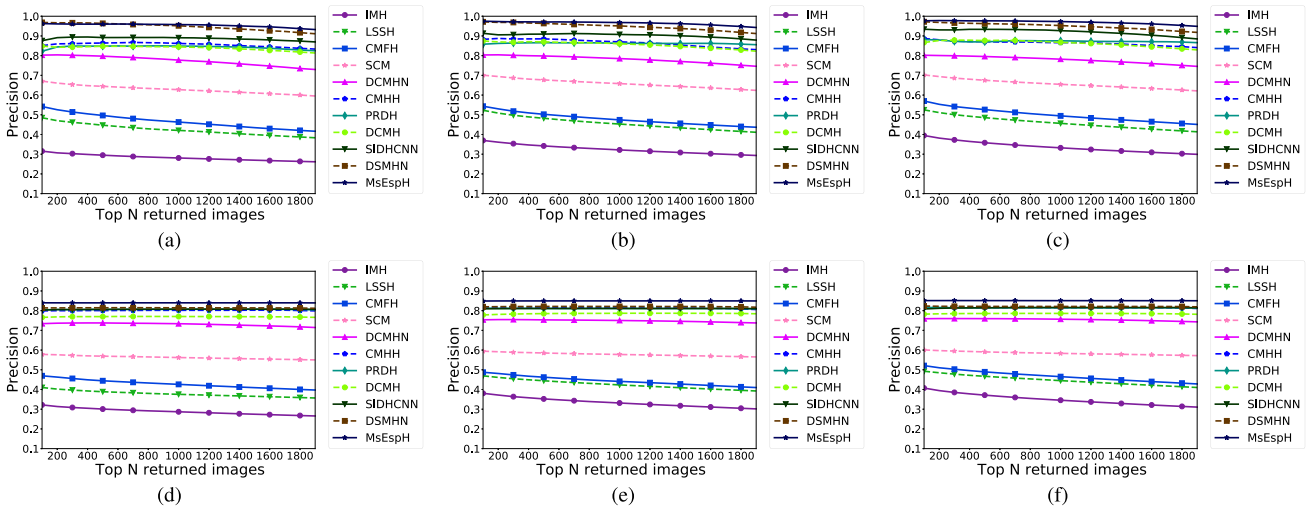


Fig. 6. P@N curve on the CBRSIR_VS dataset. (a) VHR → SAR@16 bits. (b) VHR → SAR@32 bits. (c) VHR → SAR@48 bits. (d) SAR → VHR@16 bits. (e) SAR → VHR@32 bits. (f) SAR → VHR@48 bits.

4.50%, 3.82%, 4.97%, and 4.96% increments on 16-, 24-, 32-, and 48-bit hash codes, respectively. Overall, MsEspH beats all the baselines on different lengths of hash codes on the CBRSIR_VS dataset for "VHR → SAR" and "SAR → VHR."

Fig. 5 displays the PR curves on CBRSIR_VS. Fig. 6 shows the precision with respect to different numbers of returned images (P@N) on CBRSIR_VS. From Fig. 5, we can observe that the area of the PR curves of MsEspH is bigger than all the baseline methods on 16-, 32-, and 48-bit hash codes in "VHR → SAR" and "SAR → VHR."

Meanwhile, in Fig. 6, the P@N scores of MsEspH in "VHR → SAR" and "SAR → VHR" are higher than all the baseline methods on 16-, 32-, and 48-bit hash codes. These figures demonstrate that MsEspH constantly outperforms all compared methods.

2) Qualitative Results: The VHR hashing network of MsEspH needs fuse MSIs to reduce the spatial–spectral discrepancy between VHR and SAR images. In this article, we mainly focus on the near-infrared band of MSI because, by analyzing the bad cases in CBRSIR_VS, we find that the near-infrared information can effectively distinguish some confusing RS objects or scenes. It is worth emphasizing that the network architecture of MsEspH can not only fuse MSIs but also other modal images. In this article, to obtain MSIs in real time, a pretrained Pix2Pix model is adopted. Thus, the quality of the generated near-infrared band is very important to provide accurate supplementary information for the representation of VHR images. As shown in Fig. 7, we find that the generated images by the pretrained Pix2Pix are very realistic compared with ground-truth images. Thus, these "fake" near-infrared images can be regarded as a substitute for real images when the VHR hashing network needs to fuse MSI in real time.

The performance of multisensor fusion may be decreased by some unnecessary backgrounds of RS images. Thus, we use the attention mechanism to make our model focus on key

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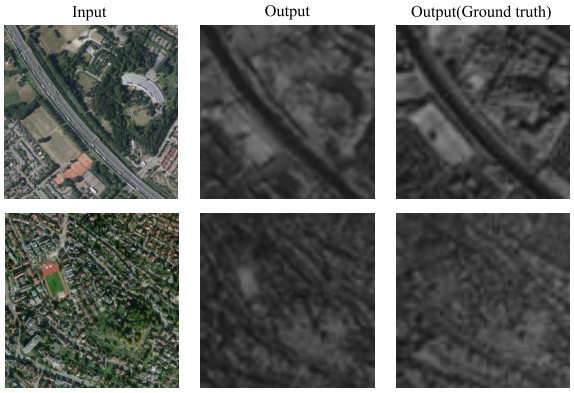


Fig. 7. Near-infrared bands of MSIs generated by a pretrained Pix2Pix.

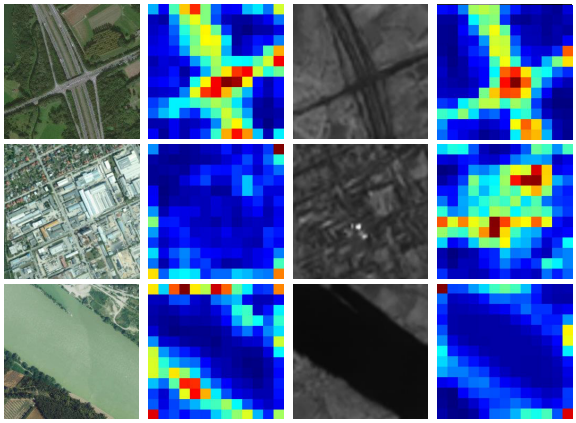


Fig. 8. Attention map of VHR images and near-infrared bands corresponding to MSIs. The first and third columns are VHR images and the near-infrared bands of MSIs, and the second and fourth columns are the corresponding attention maps.

areas, such as the outline of a highway. To check whether key areas have been learned, we visualize the learned attention scores of RS images, which is called an attention map. When the attention model is trained, the attention map can be generated based on the score functions mentioned in (4) and (5). We display the attention map by means of a pseudocolor image. As shown in Fig. 8, the first and third columns represent VHR and SAR images, and the second and fourth columns are their attention maps, respectively. The higher the brightness, the higher the attention score. From Fig. 8, we can see that the key areas can be well learned using the attention mechanism. For example, the first and third rows show that the model pays attention to the outline of a highway and the bank of a river. In particular, in the second row, the attention maps of VHR and SAR images are very different. This implies that VHR images and near-infrared bands of MSIs provide distinct spectral information, making the key areas different. These attention maps also illustrate that multisensor fusion can complement different information between VHR and SAR images to reduce the intermodal discrepancies.

When class embeddings are orthogonal in the Hamming space, i.e., the inner product is zero between each class embedding, hash learning and classification can be unified into a learning framework. Besides, as described in [58], when the binarized vectors are orthogonal, the Hamming distance between these vectors is quite large. Thus, the orthogonality of class embeddings can enhance the discrimination of generated hash codes. In the experimentation, all class embeddings are learned by a full connection layer without bias terms. To verify the orthogonality of the learned class embeddings, we visualize the inner product between the class embeddings, as shown in Fig. 9(a) and (b). The blacker the color, the smaller the inner product. Based on these two figures, we see that the inner product between most of the embeddings is close to 0. It indicates that `MsEspH` can learn the class embeddings, which is consistent with the mentioned assumptions in Section III-D. In addition, the distribution of class embeddings generally reflects the distribution of classes in the Hamming space. According to (10), the smaller the inner product between class embeddings, the larger the Hamming distance between each class. These two figures also illustrate that `MsEspH` can well learn the interclass discrimination between VHR and SAR images.

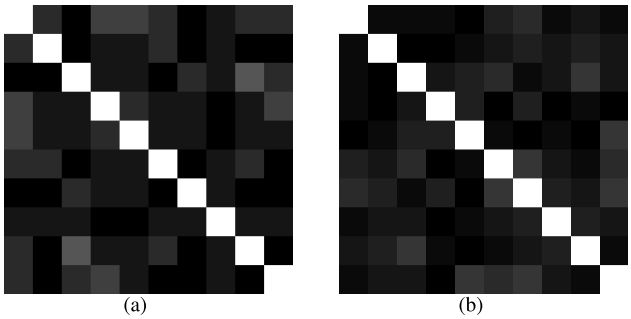


Fig. 9. Orthogonal matrices of all class embeddings on CBRSIR_VS. A darker color indicates a smaller inner product. The ith row or the ith column represents the embedding of the ith class. (a) Orthogonal matrix of all class embeddings on 24 bits. (b) Orthogonal matrix of all class embeddings on 48 bits.

3) Visualization: To compare the retrieval performance of `MsEspH` and DSMHN, we show their retrieval results on CBRSIR_VS. As shown in Fig. 10, the retrieval results of some highway and residential images are displayed. The first two queries are SAR images, and the last two queries are VHR images. As shown in Fig. 10, when retrieving highway images, including "SAR→VHR" and "VHR→SAR," DSMHN is more likely to recall wrong results. The reason behind it is that the highway images are easily confused with other images (e.g., river images) when only considering RBG bands. In contrast, `MsEspH` can well distinguish the highway and river images by fusing near-infrared information. Besides, as shown in Fig. 10, although residential and industrial images are usually not easy to be distinguished, `MsEspH` can still recall more relevant results than DSMHN. This is because interclass discrimination can be well learned by `MsEspH` through class labels directly. In summary, `MsEspH` exhibits excellent retrieval performance by fusing more spectral information and directly using class labels to guide hashing learning.

C. Performance Analysis

1) Retrieval Performance Over Training Sample Size: To examine the impact of training sample volume on

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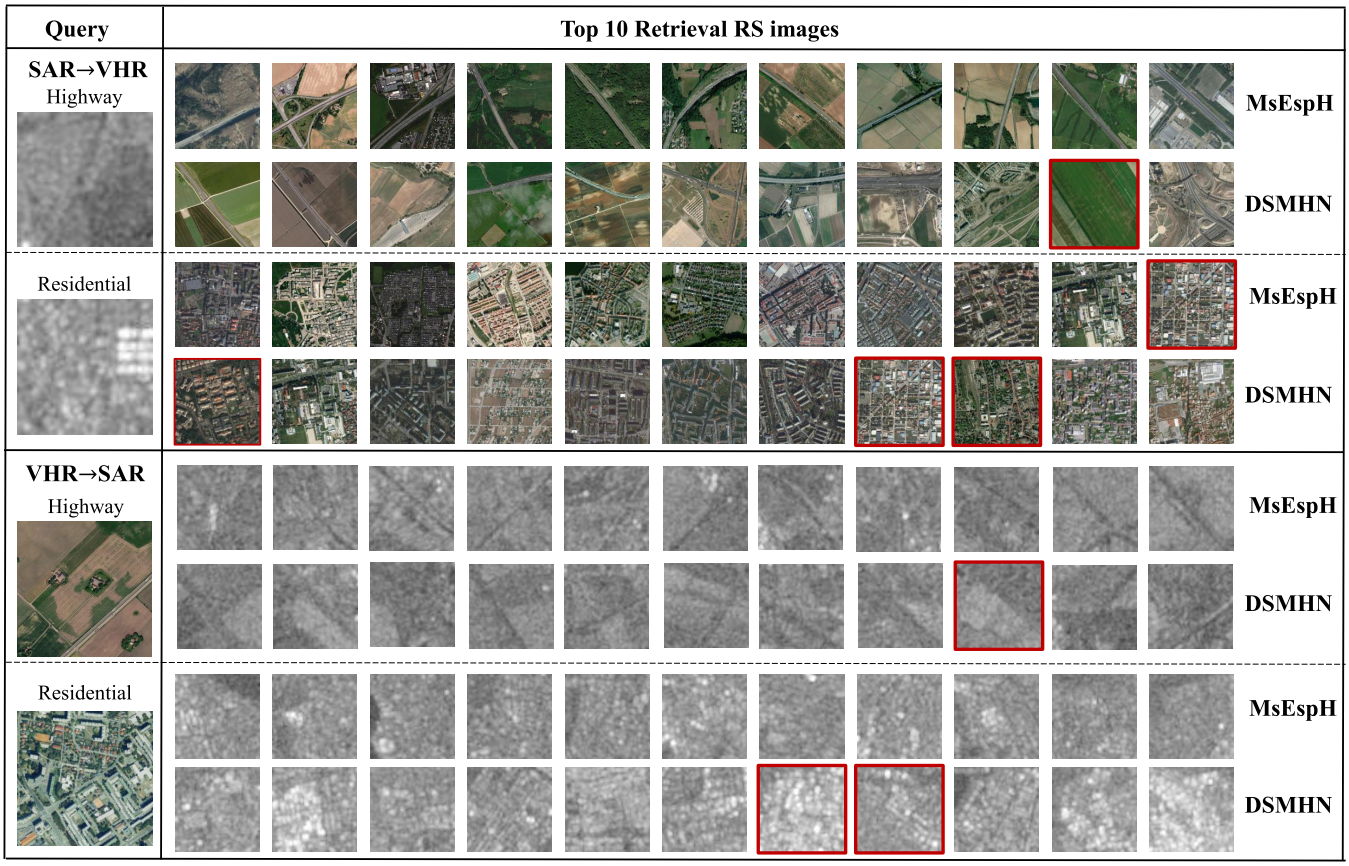


Fig. 10. Top-ten retrieved RS images by MsEspH and DSMHN on CBRSIR_VS. The retrieved RS images with red squares mean incorrect results.

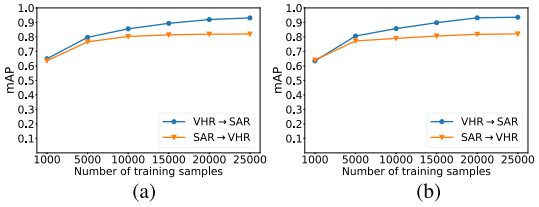


Fig. 11. Impact of the number of training samples on the performance of MsEspH. (a) Impact of the number of training samples on 24 bits. (b) Impact of the number of training samples on 48 bits.

performance, we evaluate our method with a varying size of training samples on the CBRSIR_VS dataset. Figs. 11(a) and (b) shows the results on 24 and 48 bits, respectively. In the experiment, the required number of training samples are selected randomly from the training sets, and the sample size is varied from 1000 to 25 000.

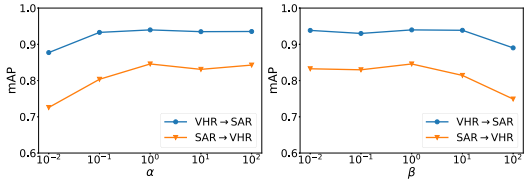
As shown in Fig. 11(a) and (b), in the beginning, the mAP scores increase rapidly as the sample size increases. After getting enough training samples, the mAP increases slowly and then gets saturated. This is because our method is a deep learning-based method, and when the sample size is too small, the deep neural network is easy to obtain poor performance [67], [68]. After the size of training data exceeds a certain value (e.g., 5000 for SAR→VHR and 15 000 for VHR→SAR), our method can obtain an excellent retrieval performance, and the mAP scores are not changed drastically as the training data increases. Therefore, our method is robust to the volume of training samples.

TABLE II
INFERENCE TIMES (MILLISECONDS) OF MsEspH ON DIFFERENT BITS FOR A VHR OR SAR QUERY IMAGE

Modality	16 bits	24 bits	32 bits	48 bits
VHR	7.45	7.51	7.38	7.19
SAR	2.11	2.20	2.15	2.00

2) Computational Efficiency Analysis: In practice, computational efficiency is very important for large-scale cross-modal RS image retrieval. In Table II, the inference times of MsEspH are reported over 16, 24, 32, and 48 bit, where the VHR and SAR denote the inference of times of the VHR hashing network and the SAR hashing network, respectively. The times are measured in milliseconds. Here, we only show the inference times of hashing networks because, in practice, the hash codes of database images can be generated offline, and the online inference time for a query image is what matters [19]. We can see that the running times of the two networks are quite small. Besides, the VHR hashing network is less efficient than the SAR hashing network. This is because the former has an additional pix2pix network and a multisensor fusion module. Nevertheless, the hash code of a VHR image can be generated in real time. Thus, our method is efficient and can be applied to large-scale cross-modal RS image retrieval.

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(a) (b)

Fig. 12. Sensitivity analysis of α and β on CBRSIR_VS. (a) mAP results of α on 48 bits. (b) mAP results of β on 48 bits.

D. Parameter Analysis

The impacts of the hyperparameters α and β are investigated over CBRSIR_VS on 48 bits. α and β control the orthogonality and quantization error between class embeddings, respectively. In our experiments, α and β are empirically set to 1 and 1. To analyze the impact of α, we fix β and vary α to evaluate the mAP of MsEspH. As shown in Fig. 12(a), we observe that the mAP score increases slowly as α increases, and the mAP decreases slightly and stabilizes basically when α is increasing beyond 10^0. This is because as α increases, the class embeddings become orthogonal, and the semantic structure of RS images can be easily learned using class labels. However, when α is large enough, the class embeddings are approximately orthogonal. At this time, the corresponding orthogonal constraint loss is close to 0, and increasing α has a limited impact on the performance of MsEspH. We adopt a similar way to evaluate β. The evaluation results are shown in Fig. 12(b), where the mAP score decreases as β increases. This is because excessive quantification may destroy the distribution of the class embeddings, i.e., the distribution of semantic structures.

In summary, α and β are used to control the discrimination and quantization error between hash codes. When α is increased in an appropriate range, the class embeddings tend to be orthogonal, i.e., the distances between them are large. According to these class embeddings, the generated hash codes have excellent discrimination. Similarly, β is varied in a suitable range, and the quantization error of MsEspH is quite small when converting continuous features into hash codes.

E. Downstream Applications

The proposed cross-modal RS image retrieval can be used for various potential applications. The cross-modal RS image retrieval can learn source-invariant feature descriptors [19], and they can be used to match dynamic similar targets on a global scale at all times. As an example, given an SAR query image captured at night, MsEspH can be used to retrieve all semantically consistent VHR images captured during daytime. Also, cross-modal RS image retrieval plays an important role in disaster rescue [2]. For example, in maritime search and rescue surveillance [21], [22], MsEspH can be applied to search similar ships in full-time and all-weather. In an emergency disaster assessment [23], MsEspH can be utilized to find more damage regions by using the image of a specific disaster region as query input.

V. Conclusion

In this article, we propose a multisensor fusion and explicit semantic preserving-based deep hashing method, called MsEspH, for VHR and SAR image retrievals. MsEspH consists of a new cross-modal hashing network and an explicit semantic preserving-based objective function. The cross-modal hashing network contains a VHR hashing network and an SAR hashing network. The VHR hashing network first uses a pretrained Pix2Pix model to generate MSI images, then fuses these images to learn the representation of VHR images, and, finally, takes a hash layer to generate the hash codes of VHR images. The SAR hashing network employs convolutional neural networks and a hash layer to produce the hash codes of SAR images. Furthermore, we design an explicit semantic preserving-based objective function to constrain cross-modal hash codes into a common semantic space. The objective function derives from a softmax function, which can explicitly preserve the intraclass and interclass similarities with class labels directly. In addition, we prove that classification and hashing learning have a certain connection and can be unified into a learning framework under some constraints. Finally, we contribute a publicly available VHR-SAR RS image dataset, namely, CBRSIR_VS. Extensive experiments are conducted on the dataset to evaluate the effectiveness of MsEspH. Compared with various state-of-the-art methods, MsEspH consistently achieves the best result.

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